

AFM (Module -2)
Capital Structure Theories

Net Income Approach

1. X Ltd. is expecting an annual EBIT of Rs.1 lakh. The company has Rs.4 lakhs in 10% debentures. The cost of equity capital is 12.5%. You are required to calculate the total value of the firm according to the Net Income Approach.

2. ABC Ltd earned a profit of Rs.20,00,000 before interest and tax. The Co's Capital structure is as follows

- a) 400,000 equity shares of Rs.10 each with a capitalization rate of 16%
- b) 25,000, 14% debentures of Rs.150 each.

You are required to compute the following using NI approach

(i) The overall cost of capital (ii) Market Value of the firm.

3. (a) A company expects a net operating income of Rs.80,000. It has ₹2,00,000, 8% Debentures. The equity capitalization rate of the company is 10%. Calculate the value of the firm and overall capitalization rate according to the Net Income Approach.

(b) If the debenture debt is increased to ₹ 6,00,000, what shall be the value of the firm and the overall capitalization rate?

Net Operating Income Approach

4. X Ltd has earned a profit of Rs.500,000 before interest and tax. The company's capital structure includes 20,000, 14% debentures of Rs 100 each. The overall capitalization rate of the firm is 16%. Compute value of the firm and equity capitalization rate using NOI approach .

5. Summer Ltd expects annual operating income of Rs.2,00,000. It has Rs.5,00,000 outstanding debt, cost of debt is 10%. If the overall capitalization rate is 12.5%, what would be the total value of the firm and the equity capitalization rate according to the NET Operating Income approach.

What will be the effect of the following on the total value of the firm and equity capitalization rate, if:

- i. The firm increases the amount of debt from Rs.5,00,000 to Rs.7,50,000 and uses the proceeds of the debt to repurchase equity shares.
- ii. The firm redeems debt of Rs.2,50,000 by issuing fresh equity shares of the same amount.

Traditional Approach

6. Compute the market value of the firm, value of shares and the average cost of capital from the following information:

	₹
Net Operating Income	2,00,000
Equity Capitalization Rate :	
(a) If the firm uses no debt	10%
(b) If the firm uses ₹4,00,000 debentures	11%
(c) If the firm uses ₹6,00,000 debenture	13%

Assume that ₹ 4,00,000 debentures can be raised at 5% rate of interest whereas ₹6,00,000 debentures can be raised at 6% rate of interest.

7. A company's current net operating income (EBIT) is ₹8,00,000. The company has ₹20 lakhs of 10% debt outstanding. Its equity capitalization rate is 15%. The company is considering to increase its debt by raising additional ₹ 10 lakhs and to utilize these funds to retire the amount of equity. However, due to increased financial risk, the cost of entire debt is likely to increase to 12% and the cost of equity it 18%.

You are required to compute the market value of the company using traditional model and also make recommendations regarding the proposal.

8. K manufacturing Ltd has a total capitalization of Rs.10,00,000 and it normally earns Rs.1,00,000 (before interest and tax) The financial manager of the firm wants to take a decision regarding the capital structure. After a study of the capital market, he gathers the following data:

Amount of debt	Interest Rate %	Equity capitalization rate %
0	-	10.00
100,000	4.0	10.50
2,00,000	4.0	11.00
3,00,000	4.5	11.60
4,00,000	5.0	12.40
5,00,000	5.5	13.50
6,00,000	6.0	16.00
7,00,000	8.0	20.00

What amount of debt should be employed by the firm if the traditional approach is held valid? Assume that corporate taxes do not exist, and that the firm always maintains its capital structure at book values.

9. A Firm is expecting a net operating income of 150 crores on assets of 1500 crores. Which are financed entirely by equity. The firm's equity capitalization rate is 10%. It is considering substituting equity by issuing debentures of 300 crore at 6%. The cost of equity at this stage is expected to increase to 10.56%. It is also on contemplating the alternative of raising 600 crores through debentures when its cost will become 7% and ESH seek a compensation of 12.5 %. Prove WACC approach.

MM Approach

10. Mid Air enterprises is an all equity company. The expected EBIT of the company is Rs.24 lakhs. If the company begins to use debt, it can borrow at the rate of 8%. This rate is independent of the amount of debt. Any money raised by selling debt would be used to retire common stock. The equity capitalization rate would be 12% when no debt is used. Assuming a debt of 1 crore, using MM model without corporate taxes

- a) Determine the firm's total market value
- b) Determine the firm's value of equity
- c) Determine the firm's leverage cost of equity.

11. X and Y are two firms identical in all aspects except that X does not use any debt in its financing, while Y has Rs.1,00,000 5% debentures in its financing. Both the firms have earnings before interest and tax of Rs.25,000 and the equity capitalization rate is 10%. Assuming the corporate tax to be 50%, calculate the value of the firm using M&M approach.

12. ABC ltd. has a net operating income of Rs.3,00,000. If equity capitalization rate is 18%. Compute the value of the firm using MM (no leverage, no tax scenario). Recompile the value of the firm under the modern approach if tax rate is 40%.

13. XYZ ltd has a net operating income of Rs.700,000. Its equity capitalization rate is 12% with a taxation rate of 30%. What would be its value? If XYZ goes in for a debt of Rs.400,000 with an interest rate of 10%, will the value of the firm change.

14. ABC ltd has a cost of debt and overall cost of capital at 9% and 12% respectively. The ratio of market value of debt to equity is 0.8. What will be the expectation of the Equity shareholders from this firm if shareholders of unlevered firm expect 14% as rate of return.

15. XYZ identified a project with 100% equity and a cost of 21%. ABC ltd. Another company, identical to XYZ ltd has a capital structure of 2:1 in debt equity ratio. Kd of this company is 14%. Compute Ke of ABC ltd and prove that WACC is constant.